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Supplementary Material: WRF Cold-Pool Sensitivities Relevant to Winter-Ozone Modelling in the Uinta Basin

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1. Secondary Controlled Comparisons

Table 1 collects the secondary process and fine-terrain feedback responses on the common station and common-AGL pipeline.

Table 1. Mean secondary controlled contrasts over the five hourly spatial medians from 1400 to 1800 UTC. CMP07 is X1 minus X0, CMP08 is X2 minus X1, and CMP06 is X9 minus X0. PBL height is scheme defined across CMP08 and is retained only with that caveat.

Metric	CMP07	CMP08	CMP06
T2 bias (°C)	+0.001	+0.37	-0.004
10 m wind absolute error (m s^{-1})	+0.001	-0.145	+0.03
Surface-pressure bias (Pa)	-0.12	+6.3	-0.77
Stability $S_{50:500}$ (K)	+0.004	+0.11	+0.04
Transport wind $U_{50:500}$ (m s^{-1})	+0.006	+0.098	-0.01
PBL height (m)	+0.01	-9.2	+0.05

Table 2 summarises the controlled terrain-source brackets without treating them as a source ranking.

Table 2. Terrain-source responses for the controlled NAM one-way brackets. CMP09 is default minus fine terrain; CMP10 and CMP11 use the deliberately coarse source as the second member.

Contrast	Terrain difference	2 m response	Scope
CMP09, default minus fine	7.9 m RMS; -55 to +60 m range	Elevation-band means 0.07–0.27 K; terrain correlation below 0.3 at 1400–1600 UTC	Small d02 response; insufficient for a terrain-source ranking
CMP10/CMP11, coarse brackets	91.9/93.1 m RMS; extrema to -462/+487 m	Slope-band means 1.2–1.5 K; terrain correlation 0.50–0.67	Response varies with domain support and diagnostic

The controlled slope-aware-radiation treatment produced negligible response at the sampled basin-floor columns but an aspect-organised skin-temperature response on steep terrain. The archived shortwave field is a flat-plane quantity and did not express the treatment, while the archive lacks the surface heat-flux terms needed to test the mechanism directly. Figure 1 retains the spatial diagnostic with those limitations.

Received:
Revised:
Accepted:
Published:

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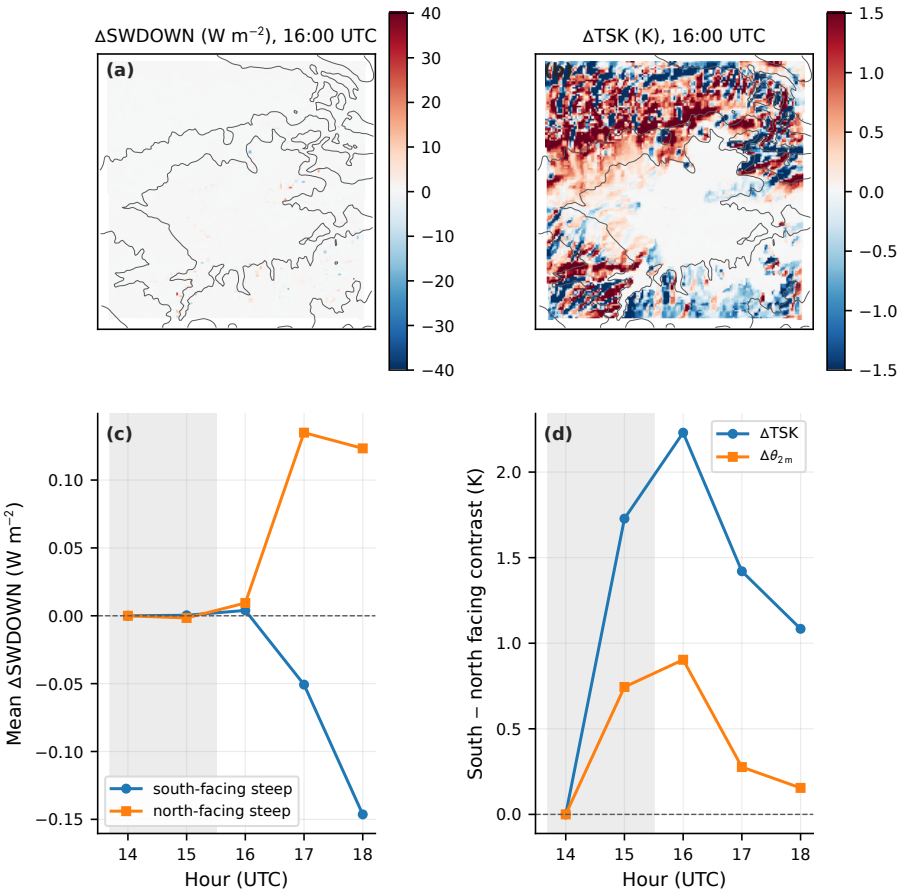


Figure 1. Diagnostic response to slope-aware radiation and terrain shading on the controlled fine-terrain branch. The archived flat-plane shortwave field showed negligible differences, whereas skin temperature showed an aspect-organised response on steep terrain. The figure demonstrates that the common station metric can miss a localised model response, but it does not show that slope-aware radiation improved or degraded the cold pool and supports no configuration ranking.

https://doi.org/10.3390/air1010000

2. Within-X8 d04 Heat-Deficit Transport

X8 is a single multiply confounded context run with 111.111 m nominal d04 grid spacing, 50 vertical levels, and 10 min output. The panels in Figure 2 describe resolved model processes within X8 and do not provide a comparison, independent validation, resolution response, numerical-viability test, or basin-export estimate.

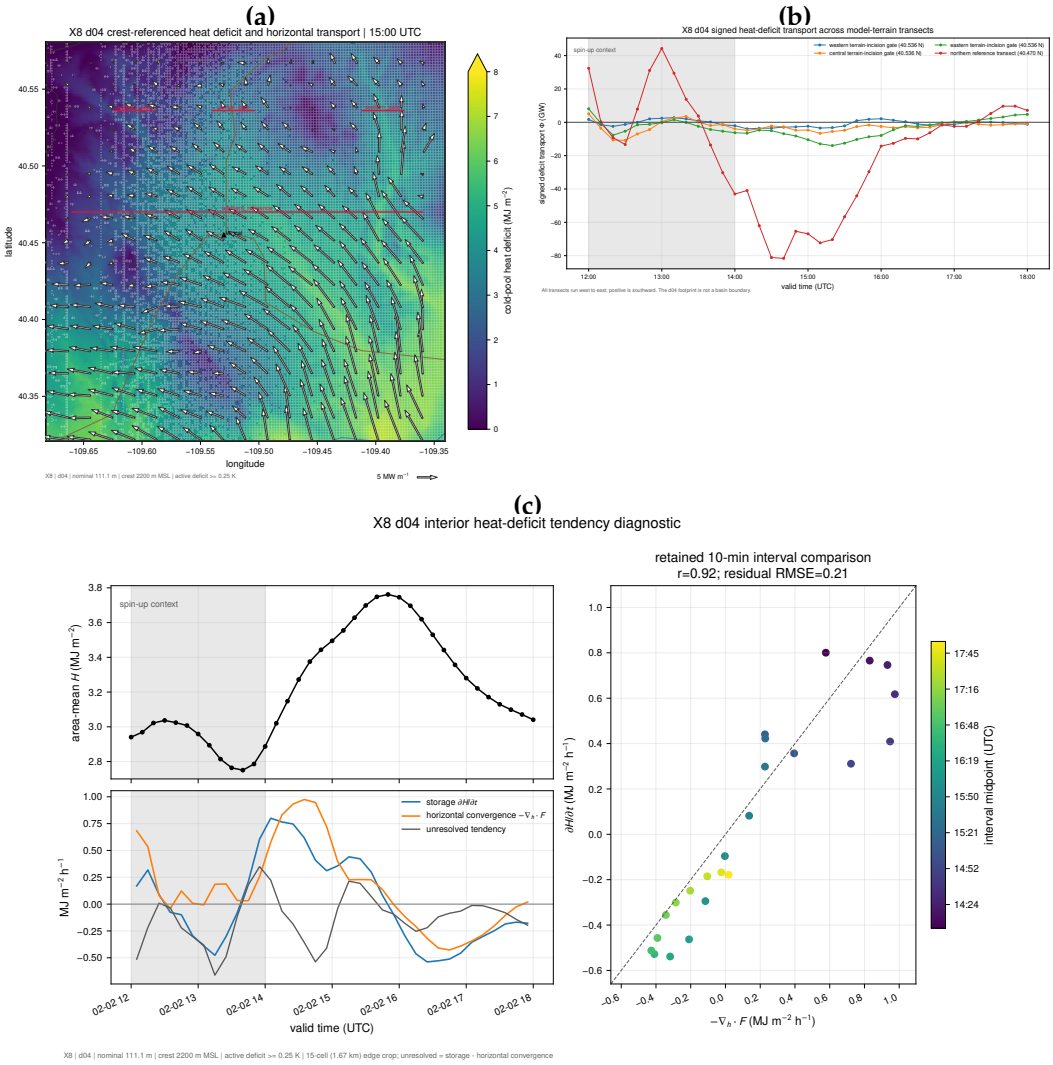


Figure 2. Within-X8 d04 heat-deficit transport and tendency diagnostics. (a) Crest-referenced heat deficit H and horizontal deficit transport $\mathbf{F}$ at 1500 UTC use the local potential temperature at 2200 m mean sea level as the column reference. The three short lines cross model-terrain incisions, while the longer line is a northern reference transect rather than a basin boundary. (b) Signed transport at 10 min cadence is positive southward and negative northward because every transect runs west to east. The northern reference line reaches -81.61 GW at 1440 UTC before relaxing to $+9.69$ GW at 1750 UTC. (c) Ten-minute finite-difference storage is compared with endpoint-mean horizontal convergence after excluding 15 cells from each d04 edge; their difference is retained as unresolved rather than attributed to a single process. Across retained intervals, storage and horizontal convergence have $r = 0.92$, while the unresolved-tendency root-mean-square error is 0.21 MJ m^{-2} h^{-1}. The shaded 1200–1400 UTC interval is spin-up context, and none of the panels establishes closed budget accounting, observed canyon drainage, or basin export.